Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

$$\text{LHS} \longrightarrow \text{RHS}$$

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

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Classified Practice **Identities and Proofs**

Q001 **Challenge**

$2x^2 + 3x + 1 = (x+1)(ax+b) + c$. **Find a,b,c.**

Classified Practice **Identities and Proofs**

Q002 **Written**

$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$

Q003 **Written**

$x^2 - x - 3 = a(x-1)^2 + b(x-1) + c$

Classified Practice **Identities and Proofs**

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x+2)^2 + b(x+2) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A $(2,\ 2,\ -3)$
B $(0,\ 2,\ -3)$
C $(1,\ 2,\ -3)$
D $(1,\ 3,\ -3)$

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x+3)^2 + b(x+3) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A $(3,\ -3,\ -1)$

B $(1,\ -3,\ -1)$

C $(2,\ -3,\ -1)$

D $(2,\ -2,\ -1)$

Classified Practice **Identities and Proofs**

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x+4)^2 + b(x+4) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (4, 4, 1)

B (2, 4, 1)

C (3, 5, 1)

D (3, 4, 1)

Q007 **MCQ**

For the identity $x^2 + 5x + 3 = a(x+5)^2 + b(x+5) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A $(1,\ -5,\ 3)$

B $(2,\ -5,\ 3)$

C $(0,\ -5,\ 3)$

D $(1,\ -4,\ 3)$

Classified Practice **Identities and Proofs**

Q008 **Written**

$a(x+1) - b(x+5) = 8x$

Q009 **MCQ**

For the identity $a(x+1) - b(x+3) = x - 1$, the ordered tuple (a, b) is:

Choose the correct option.

A (2, 1)
B (3, 1)
C (1, 1)
D (2, 2)

Classified Practice **Identities and Proofs**

Q010 MCQ

For the identity $a(x+2) - b(x+4) = 5x + 14$, the ordered tuple (a, b) is:

Choose the correct option.

A $(4,\ -2)$
B $(3,\ -2)$
C $(2,\ -2)$
D $(3,\ -1)$

Q011 **MCQ**

For the identity $a(x+3) - b(x+5) = x - 3$, the ordered tuple (a, b) is:

Choose the correct option.

A $(5,\ 3)$
B $(3,\ 3)$
C $(4,\ 4)$
D $(4,\ 3)$

Classified Practice **Identities and Proofs**

Q012 MCQ

For the identity $a(x+4)-b(x+6)=9x+44$, the ordered tuple (a,b) is:

Choose the correct option.

A $(6,\ -4)$
B $(4,\ -4)$
C $(5,\ -3)$
D $(5,\ -4)$

Q013 **Written**

$(x + a)(b + 3) = 4x + 12$

Classified Practice **Identities and Proofs**

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Choose the correct option.

A $(3,\ -1)$

B $(2,\ -1)$

C $(1,\ -1)$

D $(2,\ 0)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Choose the correct option.

A (4, 0)
B (2, 0)
C (3, 0)
D (3, 1)

Classified Practice **Identities and Proofs**

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Choose the correct option.

A $(5,\ 1)$
B $(3,\ 1)$
C $(4,\ 1)$
D $(4,\ 2)$

Q017 **Written**

$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$

Classified Practice **Identities and Proofs**

Q018 **Written**

$(ax + b)(x + 1) = 4x^2 + cx + 1$

Q019 **Written**

$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$

Classified Practice **Identities and Proofs**

Q020 MCQ

For the identity $(ax-1)(x+1)+c=x^2-3$, the ordered tuple (a,c) is:

Choose the correct option.

A $(2,\ -2)$
B $(1,\ -2)$
C $(0,\ -2)$
D $(1,\ -1)$

Q021 **MCQ**

For the identity $(ax+2)(x+2)+c=2x^2+6x+3$, the ordered tuple (a,c) is:

Choose the correct option.

A $(3,\ -1)$
B $(1,\ -1)$
C $(2,\ -1)$
D $(2,\ 0)$

Classified Practice **Identities and Proofs**

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Choose the correct option.

A $(4,\ 0)$
B $(3,\ 0)$
C $(2,\ 0)$
D $(3,\ 1)$

Q023 **Written**

$$\frac{3x+1}{(x+2)(x-3)} = \frac{a}{x+2} + \frac{b}{x-3}$$

Classified Practice **Identities and Proofs**

Q024 **Written**

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Q025 **Written**

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Classified Practice　　**Identities and Proofs**

Q026 **Written**

$$\frac{a}{x-1}+\frac{b}{x-5}=\frac{5x-1}{(x-1)(x-5)}$$

Q027 **Written**

$$\frac{3x-8}{3x^2+5x-2} = \frac{a}{x+2} + \frac{b}{3x-1}$$

Classified Practice **Identities and Proofs**

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3},\ t > 0$$

Q029 **Written**

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.

Classified Practice **Identities and Proofs**

Q030 **Written**

Prove that $(2a+b)^2-(a+2b)^2=3\left(a^2-b^2\right)$.

Q031 MCQ

For the identity $-(x-2y+1)^2+(x+2y+1)^2=2y(4x+4)$**, the value of the simplified left-hand side is:**

Choose the correct option.

A $8xy+8y$

B $8xy+8y+1$

C $8xy+8y-1$

D $-8xy-8y$

Classified Practice **Identities and Proofs**

Q032 MCQ

For the identity $-(x-3y+2)^2+(x+3y+2)^2=3y(4x+8)$, the value of the simplified left-hand side is:

Choose the correct option.

A $12xy+24y+1$

B $12xy+24y-1$

C $-12xy-24y$

D $12xy+24y$

Q033 MCQ

For the identity $-(x-y+3)^2+(x+y+3)^2=y(4x+12)$**, the value of the simplified left-hand side is:**

Choose the correct option.

A $4xy+12y+1$

B $4xy+12y-1$

C $-4xy-12y$

D $4xy+12y$

Classified Practice **Identities and Proofs**

Q034 MCQ

For the identity $-(x-2y+4)^2+(x+2y+4)^2=2y(4x+16)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8xy+32y$

B $8xy+32y+1$

C $8xy+32y-1$

D $-8xy-32y$

Q035 **Written**

Prove that $(x+y)^2+(x-y)^2=2x^2+2y^2$.

Classified Practice **Identities and Proofs**

Q036 MCQ

For the identity $(x-2y+1)^2+(x+2y+1)^2=8y^2+2(x+1)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2+4x+8y^2+3$

B $2x^2+4x+8y^2+1$

C $2x^2+4x+8y^2+2$

D $-2x^2-4x-8y^2-2$

Q037 MCQ

For the identity $(x-3y+2)^2+(x+3y+2)^2=18y^2+2(x+2)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2+8x+18y^2+9$

B $2x^2+8x+18y^2+7$

C $2x^2+8x+18y^2+8$

D $-2x^2-8x-18y^2-8$

Classified Practice **Identities and Proofs**

Q038 MCQ

For the identity $(x-y+3)^2+(x+y+3)^2=2y^2+2(x+3)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2+12x+2y^2+19$

B $2x^2+12x+2y^2+17$

C $2x^2+12x+2y^2+18$

D $-2x^2-12x-2y^2-18$

Q039 **MCQ**

For the identity $(x-2y+4)^2+(x+2y+4)^2=8y^2+2(x+4)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2+16x+8y^2+33$

B $2x^2+16x+8y^2+31$

C $-2x^2-16x-8y^2-32$

D $2x^2+16x+8y^2+32$

Classified Practice **Identities and Proofs**

Q040 **Written**

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.

Q041 MCQ

For the identity $-(x-2y+3)^2+(x+2y-1)^2=(8x+8)(y-1)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8xy-8x+8y-7$

B $8xy-8x+8y-9$

C $-8xy+8x-8y+8$

D $8xy-8x+8y-8$

Classified Practice **Identities and Proofs**

Q042 MCQ

For the identity $-(2x-3y+5)^2+(2x+3y-1)^2=(24x+24)(y-1)$**, the value of the simplified left-hand side is:**

Choose the correct option.

A $24xy-24x+24y-23$

B $24xy-24x+24y-25$

C $24xy-24x+24y-24$

D $-24xy+24x-24y+24$

Q043 **Written**

Prove that $\left(x^2 + x + 1\right)\left(x^2 - x + 1\right) = x^4 + x^2 + 1.$

Classified Practice **Identities and Proofs**

Q044 Written

Prove that $\left(a^2 - ab + b^2\right)\left(a^2 + ab + b^2\right) = \left(a^2 + b^2\right)^2 - (ab)^2$.

Q045 **Written**

Prove that $\left(a^2+b^2\right)\left(c^2+d^2\right)=(ac-bd)^2+(ad+bc)^2$.

Classified Practice **Identities and Proofs**

Q046 **Written**

Prove that $\left(x^2+1\right)\left(y^2+1\right)=(xy+1)^2+(x-y)^2$.

Q047 **Written**

Prove that $\left(a^2 - b^2\right)\left(c^2 - d^2\right) = (ac - bd)^2 - (ad - bc)^2$.

Classified Practice **Identities and Proofs**

Q048 **Written**

Prove that $(a+b)^3 - 3ab\,(a+b) = a^3 + b^3$.

Q049 **Written**

Prove that $(a + b + c)\left(a^2 + b^2 + c^2 - ab - bc - ca\right) = a^3 + b^3 + c^3 - 3abc$.

Classified Practice **Identities and Proofs**

Q050 Challenge

For the integrated identity $\left(x^2+1\right)\left(-\left(x-2y+1\right)^2+\left(x+2y+1\right)^2\right)=2y\left(4x+4\right)\left(x^2+1\right)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8x^3y+8x^2y+8xy+8y+1$

B $8x^3y+8x^2y+8xy+8y$

C $8x^3y+8x^2y+8xy+8y-1$

D $-8x^3y-8x^2y-8xy-8y$

Classified Practice **Identities and Proofs**

Q051 **Written**

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Q052 **Written**

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Classified Practice **Identities and Proofs**

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2xy + 5$

B $-2xy + 3$

C $-2(xy - 2)$

D $2(xy - 2)$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 5)$
B $-2(xy - 4)$
C $2xy - 9$
D $-2xy + 9$

Classified Practice **Identities and Proofs**

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 1)$

B $-2xy$

C $-2xy + 1$

D $2xy - 1$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 8)$

B $-2xy + 17$

C $-2xy + 15$

D $2(xy - 8)$

Classified Practice **Identities and Proofs**

Q057 **Written**

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2(xy + 1)$

B $2xy + 1$

C $2xy$

D $-2xy - 1$

Classified Practice **Identities and Proofs**

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2xy + 5$

B $2(xy + 2)$

C $2xy + 3$

D $-2(xy + 2)$

Q060 MCQ

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2xy + 5$

B $2xy + 3$

C $2(xy + 2)$

D $-2(xy + 2)$

Classified Practice **Identities and Proofs**

Q061 **MCQ**

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2\,(xy + 5)$
B $2\,(xy + 4)$
C $-2xy - 9$
D $2xy + 9$

Q062 **Written**

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Classified Practice **Identities and Proofs**

Q063 **Written**

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Q064 **Written**

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Classified Practice **Identities and Proofs**

Q065 **Written**

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Q066 **Written**

Under the condition $a + b + c = 0$, **prove that** $(a + b)(b + c)(c + a) + abc = 0$.

Classified Practice **Identities and Proofs**

Q067 **Written**

Under the condition $a + b + c = 0, \quad abc \neq 0$**, prove that** $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Classified Practice **Identities and Proofs**

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Choose the correct option.

A $-2xy + 11$

B $-2xy + 9$

C $-2(xy - 5)$

D $2(xy - 5)$

Classified Practice **Identities and Proofs**

Q069 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice **Identities and Proofs**

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\dfrac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\dfrac{cd}{c^2 + d^2}$

D $\dfrac{cd}{c^2 + d^2}$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice **Identities and Proofs**

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\dfrac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\dfrac{cd}{c^2 + d^2}$

D $\dfrac{cd}{c^2 + d^2}$

Classified Practice **Identities and Proofs**

Q074 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\dfrac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\dfrac{cd}{c^2 + d^2}$

D $\dfrac{cd}{c^2 + d^2}$

Classified Practice **Identities and Proofs**

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice **Identities and Proofs**

Q076 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\quad \frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Classified Practice **Identities and Proofs**

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{a^2+c^2}{(a-c)(a+c)}$

B $\dfrac{2a^2}{(a-c)(a+c)}$

C $\dfrac{2c^2}{(a-c)(a+c)}$

D $-\dfrac{a^2+c^2}{(a-c)(a+c)}$

Classified Practice **Identities and Proofs**

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{2a^2}{(a-c)(a+c)}$

B $\dfrac{2c^2}{(a-c)(a+c)}$

C $-\dfrac{a^2+c^2}{(a-c)(a+c)}$

D $\dfrac{a^2+c^2}{(a-c)(a+c)}$

Classified Practice **Identities and Proofs**

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{2a^2}{(a-c)(a+c)}$

B $\dfrac{2c^2}{(a-c)(a+c)}$

C $-\dfrac{a^2+c^2}{(a-c)(a+c)}$

D $\dfrac{a^2+c^2}{(a-c)(a+c)}$

Classified Practice **Identities and Proofs**

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{a^2+c^2}{(a-c)(a+c)}$

B $\dfrac{2a^2}{(a-c)(a+c)}$

C $\dfrac{2c^2}{(a-c)(a+c)}$

D $-\dfrac{a^2+c^2}{(a-c)(a+c)}$

Classified Practice **Identities and Proofs**

Q081 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{2c}{c-d}$

B $\frac{c+d}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

Classified Practice **Identities and Proofs**

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{2c}{c-d}$

B $\dfrac{c+d}{c-d}$

C $\dfrac{2d}{c-d}$

D $-\dfrac{c+d}{c-d}$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{c+d}{c-d}$

B $\dfrac{2c}{c-d}$

C $\dfrac{2d}{c-d}$

D $-\dfrac{c+d}{c-d}$

Classified Practice **Identities and Proofs**

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{c+d}{c-d}$

B $\dfrac{2c}{c-d}$

C $\dfrac{2d}{c-d}$

D $-\dfrac{c+d}{c-d}$

Q086 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Classified Practice **Identities and Proofs**

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\dfrac{2a-c}{2b-d}$

B $\dfrac{2a+2b-c-d}{2b-d}$

C $\dfrac{2a-2b-c+d}{2b-d}$

D $-\dfrac{2a-c}{2b-d}$

Q088 **Written**

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Classified Practice **Identities and Proofs**

Q089 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q090 **Written**

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\quad \frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$.

Classified Practice **Identities and Proofs**

Q091 **Written**

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Q092 **Written**

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Classified Practice **Identities and Proofs**

Q093 **Written**

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Q094 **Written**

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Classified Practice **Identities and Proofs**

Q095 **Written**

Under the condition $a : b : c = 3 : 5 : 2$**, prove or find the required expression:** $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Classified Practice **Identities and Proofs**

Q097 **Written**

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0, \quad \frac{x^2+5y^2}{x^2+xy+y^2}$.

Q098 **Written**

Under the condition $x/a = y/b = z/c$**, prove or find the required expression:** $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Classified Practice **Identities and Proofs**

Quiz MCQ 1 Concept Quiz · MCQ

If $x^2 + 4x + 1 = (x+1)(ax+b) + c$, then (a, b, c) is... Which of the following is correct?:

Choose the correct option.

A $(1, 3, -2)$
B $(1, 2, -1)$
C $(2, 3, -2)$
D $(1, -3, 2)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For the identity $\left(x^2-2x+2\right)\left(x^2+2x+2\right)=-4x^2+\left(x^2+2\right)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A x^4+4

B x^4+5

C x^4+3

D $-x^4-4$

Classified Practice **Identities and Proofs**

Quiz MCQ 3 Concept Quiz · MCQ

Under the condition $a+b+c=0$, the expression $3a^2+3ac$ is equal to:

Choose the correct option.

A $3b\,(b+c)$
B $3b^2+3bc+1$
C $3b^2+3bc-1$
D $-3b\,(b+c)$

Quiz MCQ 4 **Concept Quiz · MCQ**

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Choose the correct option.

A $\frac{5}{2}$

B $\frac{3}{2}$

C $\frac{1}{2}$

D $-\frac{3}{2}$

Classified Practice **Identities and Proofs**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find a, b, c **if** $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Prove that $\left(x^2 - 3x + 4\right)\left(x^2 + 3x + 4\right) = -9x^2 + \left(x^2 + 4\right)^2$.

Classified Practice **Identities and Proofs**

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Quiz WRITTEN 8 **Concept Quiz · Written**

Under the condition $x/a = y/b = z/c$, prove that $\left(a^2 + b^2 + c^2\right)\left(x^2 + y^2 + z^2\right) = (ax + by + cz)^2$.

Answer key

Use the question number shown on each page.

Q001 $a = 2,\ b = 1,\ c = 0$

Q002 $a = 1,\ b = -2,\ c = 7$

Q003 $a = 1,\ b = 1,\ c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10,\ b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3,\ b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2,\ b = 3,\ c = 1$

Q018 $a = 4,\ b = 1,\ c = 5$

Q019 $a = 2,\ b = -1,\ c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1,\ b = 2$

Q024 $a = 2,\ b = -1$

Q025 $a = \frac{2}{3},\ b = -\frac{1}{3}$

Q026 $a = -1,\ b = 6$

Q027 $a = 2,\ b = -3$

Q028 $a = 4,\ b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} = -3$

Q068 C

Q069 $\frac{ab}{a^2 + b^2} = \frac{cd}{c^2 + d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a + b}{a - b} = \frac{c + d}{c - d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a + c}{3b + d} = \frac{2a - c}{2b - d}$

Q087 A

Q088 $\frac{a^2 + x^2}{a^2 - x^2} = \frac{b^2 + y^2}{b^2 - y^2}$

Q089 $\frac{a + b}{a - b} = \frac{c + d}{c - d}$

Q090 $\frac{x^2 + y^2 + z^2}{a^2 + b^2 + c^2} = \frac{xy + yz + zx}{ab + bc + ca}$

Q091 $\frac{x + y + z}{a + b + c} = \frac{x}{a}$

Q092 $\frac{x + 2y + 3z}{a + 2b + 3c} = \frac{x + y + z}{a + b + c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a = 1,\ b = -1,\ c = 3$

Quiz WRITTEN 6 $x^4 - x^2 + 16$

Quiz WRITTEN 7 $4b^2 + 4bc$

Quiz WRITTEN 8 $(ax + by + cz)^2$